

CHAPTER 5 — String Library Functions

Subtopics

1. `strcpy()`
 2. `strcat()`
 3. `strcmp()`
 4. `strchr()`
 5. `<string.h>` Library
-

SLIDE 1 — `strcpy()`

What is `strcpy()`?

`strcpy()` is used to **copy one string into another string**.

Think of it like copying a message from one notebook  to another.

Source  Destination 
"Hello" —————→ "Hello"

Example

```
#include <stdio.h>
#include <string.h>
```

```
int main()
{
    char source[] = "Hello";
    char destination[20];

    strcpy(destination, source);
```

```
printf("%s", destination);  
  
return 0;  
}
```

Output:

Hello

What is happening?

source → "Hello"



strcpy()



destination → "Hello"

Remember:

strcpy() = String COPY 

SLIDE 2 — strcat()

What is strcat()?

strcat() is used to join two strings together.

Think of it like connecting two train compartments  + .

"Hello" + "World"



"HelloWorld"

Example

```
#include <stdio.h>  
#include <string.h>
```

```
int main()
{
    char first[30] = "Hello ";
    char second[] = "World";

    strcat(first, second);

    printf("%s", first);

    return 0;
}
```

Output:

Hello World

What is happening?

first → "Hello "

second → "World"



strcat()



first → "Hello World"

Memory Trick:

`strcat()` = String CATenate → JOIN 

SLIDE 3 — `strcmp()`

What is `strcmp()`?

`strcmp()` is used to **compare two strings**.

Think of it like checking two answers and asking:

“Are these strings the same?” 🤖

Example

```
#include <stdio.h>
#include <string.h>

int main()
{
    char first[] = "Hello";
    char second[] = "Hello";

    if(strcmp(first, second) == 0)
    {
        printf("Strings are same");
    }

    return 0;
}
```

Output:

Strings are same

What is happening?

first → "Hello"
second → "Hello"
↓
strcmp()
↓
Compare ⚖️
↓
Same ✅

For beginners, remember:

strcmp() == 0 → Strings are equal ✅

Memory Trick:

`strcmp()` = String COMPARE ⚖️

🟠 SLIDE 4 — `strchr()` 🔍

📌 What is `strchr()`?

`strchr()` is used to **search for a particular character inside a string**.

Think of it like using a magnifying glass 🔍 to find one letter.

Example:

"HELLO"

H → ❌

E → ❌

L → ✅ Found!

💻 Example

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char word[] = "HELLO";
```

```
    if(strchr(word, 'L') != NULL)
```

```
    {
```

```
        printf("Character found");
```

```
    }
```

```
    return 0;
```

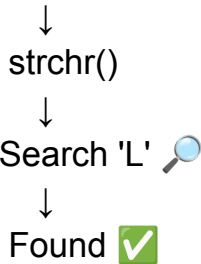
```
}
```

Output:

Character found

What is happening?

word → "HELLO"



Memory Trick:

`strchr()` = String CHaRacter search 

SLIDE 5 — `<string.h>` Library + Recap

What is `<string.h>`?

`<string.h>` is a **library header file** that provides useful functions for working with strings.

To use these string functions, we include:

```
#include <string.h>
```

Our String Toolkit

Function	What it does
----------	--------------

<code>strcpy()</code>	Copies a string
-----------------------	-----------------

<code>strcat()</code>	Joins two strings
-----------------------	-------------------

`strcmp()` Compares two strings

`strchr()` Searches for a character

BIG MEMORY TRICK

COPY → JOIN → COMPARE → SEARCH

 `strcpy()`



 `strcat()`



 `strcmp()`



 `strchr()`

Quick Recap

 `strcpy()` → Copy

 `strcat()` → Join

 `strcmp()` → Compare

 `strchr()` → Search

 `<string.h>` → Provides string functions

LEVEL CLEARED!

You can now use C's basic **string toolkit** to copy, join, compare, and search strings.

Questions:-

Q1. MCQ — strcpy()

Which function is used to **copy one string into another string**?

- ♦ A) strcat()
- ♦ B) strcmp()
- ♦ C) strcpy()
- ♦ D) strchr()

 **Hint:** Think **COPY** .

 **Glossary:**

-  **Copy** — Make another string with the same characters.

 **Answer: C) strcpy()**

Q2. MCQ — strcat()

What is the purpose of `strcat()`? 

- ♦ A) Compare two strings
- ♦ B) Join two strings
- ♦ C) Search for a character
- ♦ D) Find string length

 **Hint:** Think of `strcat()` as **connecting** two strings.

 **Glossary:**

-  **Join** — Combine two things together.

 **Answer: B) Join two strings**

Q3. Fill in the Blank — strcpy()

Complete the code to copy `source` into `destination`.

```
#include <stdio.h>
#include <string.h>

int main()
{
    char source[] = "Hello";
    char destination[20];

    _____(destination, source);

    printf("%s", destination);

    return 0;
}
```

 **Hint:** Which function performs a **string copy**?

Glossary:

-  **Copy** — Make a duplicate of the string.
-  **Destination** — The string where the copied value goes.
-  **Source** — The string from which the value is copied.

 **Answer:** `strcpy()`

Q4. Fill in the Blank — `strcat()`

Complete the code to join the two strings.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char first[30] = "Hello ";
    char second[] = "World";

    _____(first, second);
}
```

```
printf("%s", first);

return 0;
}
```

 **Hint:** Which function is used to **join strings together**?

Glossary:

-  **Join** — Combine two strings.
-  **String** — A sequence of characters.

 **Answer:** `strcat()`

Q5. Code Choice — `strcmp()`

What will be the output of this program? 

```
#include <stdio.h>
#include <string.h>

int main()
{
    char first[] = "CAT";
    char second[] = "CAT";
    if(strcmp(first, second) == 0)
        printf("Same");
    else
        printf("Different");
    return 0;
}
```

- ♦ A) **Same** 
- ♦ B) **Different**
- ♦ C) **CAT**
- ♦ D) **No output**

💡 **Hint:** `strcmp()` compares two strings. When the strings are equal, it returns 0.

📖 **Glossary:**

- ⚖️ **Compare** — Check two strings against each other.
- ✅ **Same** — Equal or matching.

Answer: A) Same